A PRODUCT OF VALUATION RINGS INSIDE A FIELD IS A GENERALIZED VALUATION RING OF WEAK DIMENSION ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- For an element of a field not lying in a subring there is a valuation ring of the field containing the subring and not containing that element. invoked at proof lines 30692-30693. Supplies the family of valuation rings whose product is built in the proof.
- A product of rings of weak dimension at most one has weak dimension at most one, and the same holds for the ring appearing as the product of the fields. invoked at proof lines 30695-30697. Gives the weak dimension statement for the constructed ring, in both of the forms the statement asks about.

Source reference. Original lemma 090P: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 30692-30693 to choose the valuation ring attached to each element outside the subring; the product construction built on that choice is inside the counted span.

Source reference. Original lemma 092T: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30648–30648. Invoked at proof lines 30695-30697 for the weak dimension conclusions in both forms; the identification of the constructed rings with the ones the statement names is inside the counted span.

Source reference. Original lemma 00HT: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 30699-30701 for the flatness of the localization map V to L; the construction of V and the weak dimension statements are inside the counted span.

Source reference. Original lemma 04VS: sources/stacks-chow/tags.txt, source lines 1–1. Cited together with the flatness statement at proof lines 30699-30701 for the two properties of the localization map; the argument using those properties is inside the counted span.

Lemma 0.2. *Let A be a normal domain with fraction field K. There exists a cartesian diagram*

$$\begin{array}{ccc} A & \longrightarrow & K \\ \downarrow & & \downarrow \\ V & \longrightarrow & L \end{array}$$

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

of rings where V has weak dimension at most 1 and $V \to L$ is a flat, injective, epimorphism of rings.

Proof. For every $x \in K$, $x \notin A$ pick $V_x \subset K$ as in Algebra, Lemma 090P. Set $V = \prod_{x \in K \setminus A} V_x$ and $L = \prod_{x \in K \setminus A} K$. The ring V has weak dimension at most 1 by Lemma 092T which also shows that $V \to L$ is a localization. A localization is flat and an epimorphism, see Algebra, Lemmas 00HT and 04VS. □